SL Stomata Labs

Working Protocols

Organisation Structure, Roles & Operating Logic

A joint venture between Pantaleon and Findability Sciences

Version 2.0 · Confidential — Internal use only

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1. Introduction

Stomata Labs is a joint venture between **Pantaleon**, a leading sugar mill group with deep operational expertise in sugarcane agriculture, industrial processing, and commercial logistics, and **Findability Sciences**, an applied AI company specialising in machine learning, data engineering, and software product development. The venture exists to build and commercialise the Stoma platform — a five-layer suite of analytics and AI products that transforms how sugar mills make decisions.

This document describes how Stomata Labs is organised to deliver on that mission. It explains the logic behind the organisational structure, defines every role, and describes how the different functions interact operationally. It is intended for all team members — technical and commercial — as well as the Steering Committee.

This is version 2.0, updated to reflect the full nine-role Build team, the addition of Stoma Sense as the fifth product layer, and the expanded working protocol set. A companion interactive HTML guide provides a searchable, filterable reference for all working protocols.

→ **Note for Steering Committee.** This document is the foundation for Stomata Labs' OKR process. Every role described here maps to a set of measurable objectives. Section 8 explains how to use this structure to set OKRs at the team and individual level.

2. Why this organisation

The structure of Stomata Labs was designed from first principles, starting with a simple question: *what does this venture need to do in order to succeed?* The answer has three non-negotiable components.

2.1 The Build / Sell / Keep framework

Any B2B software venture can fail in exactly three ways: it fails to build a product that works, it fails to sell that product to customers, or it fails to keep customers after they have signed. These are not sequential risks — they operate simultaneously from the first day. The organisational structure maps directly onto these three failure modes.

BUILD
Make the product work

- Define the product roadmap
- Develop and validate AI/ML models
- Build Sense geospatial intelligence
- Build data infrastructure (Terra)
- Create dashboards and UX (Basic)
- Develop anomaly detection (Prime)
- Build digital twins and prescriptions (Nexus)

SELL
Bring in new accounts

- Generate and manage the pipeline
- Lead commercial negotiations
- Deploy domain expertise in pre-sales
- Own pricing and packaging strategy
- Feed market intelligence to product
- Manage competitive monitoring

KEEP
Retain and grow accounts

- Own post-signature relationships
- Run layered onboarding (5 layers)
- Drive operator adoption and change
- Manage renewals and expansions
- Feed field insights back to Build
- Hold veto over expansion timing

2.2 Why each pillar matters for Stomata

Build matters because the product is technically ambitious. The Stoma platform spans five layers of increasing complexity — from data infrastructure through descriptive analytics, predictive AI, field intelligence, and prescriptive optimisation. Building this requires engineers with industrial AI expertise, geospatial and remote sensing capability, full-stack development skills, MLOps discipline, and domain knowledge of sugar mill processes — all working in an integrated way under a single product lead.

Sell matters because the market is specialised. The global sugar mill industry is a tight-knit community of technically sophisticated buyers. A standard enterprise software sales motion does not work here. Every conversation requires domain credibility — the ability to discuss POL optimisation, extraction efficiency, crystallisation dynamics, or satellite-derived crop health at the level of a process engineer or chief agronomist.

Keep matters because sugar mills are inherently conservative. Industrial operators do not change how they make decisions quickly. Signing a contract does not mean adoption. A mill that has installed the product but does not use it is a churn risk. The Keep pillar exists to close the gap between deployment and genuine adoption across all five product layers — which requires a structured change management capability, not just account management.

■ The three pillars are not independent. A Build team that does not receive field feedback from Keep will build the wrong things. A Sell team that promises capabilities Build cannot deliver will create churned accounts. The working protocols in this document exist to ensure information flows correctly between all three pillars at all times.

3. Product architecture — the five Stoma layers

The Stoma platform is a progressive intelligence stack. Each layer answers a different operational question and builds on the layer below it. The commercial strategy is built around a land-and-expand motion that moves accounts progressively through the stack.

STOMA NEXUS Prescriptive — What needs to be done?	Digital twins · world models · optimisation · LLM copilots
STOMA SENSE Intelligence — What is happening in the field?	Satellite data · remote sensing · field intelligence · AI models
STOMA PRIME Predictive — What will happen?	Anomaly detection · time/frequency domain · reconstruction error
STOMA BASIC Descriptive — What is happening?	Dashboards · SPC charts · trend views · real-time alerts
STOMA TERRA Integration — How does data get here?	IoT sensors · satellite · climate APIs · cloud data lake

STOMA TERRA

The foundational data infrastructure layer. Connects IoT sensors on the factory floor and in the field, ingests satellite data and climate API feeds, and consolidates everything into a unified cloud data lake. Terra is a prerequisite for every other layer. The Data/Cloud Engineer owns the architecture and quality SLAs.

STOMA BASIC

The descriptive analytics layer, answering: what has happened, and what is happening right now? Online dashboards, SPC charts, trend views, and real-time alerts. The recommended land product — fastest to deploy, least organisational change required, builds the data trust that all upper layers depend on. Owned by the Full Stack Developer.

STOMA PRIME

The predictive analytics layer, answering: what will happen? Applies industrial AI methods — time and frequency domain analysis, autoencoders, PCA-based reconstruction error, cross-variable correlation — to detect anomalies before they become failures. Owned by the AI/ML Engineer. Prime quality depends entirely on Terra data quality.

STOMA SENSE

The geospatial intelligence layer, answering: what is happening in the field? Integrates satellite data, remote sensing indices (NDVI, EVI, LAI), and AI models to deliver field intelligence covering crop health, variability mapping, and agronomic insights. Owned by the Geospatial/Remote Sensing Engineer. Sense is a committed product layer with a distinct onboarding journey requiring field calibration and agronomic interpretation.

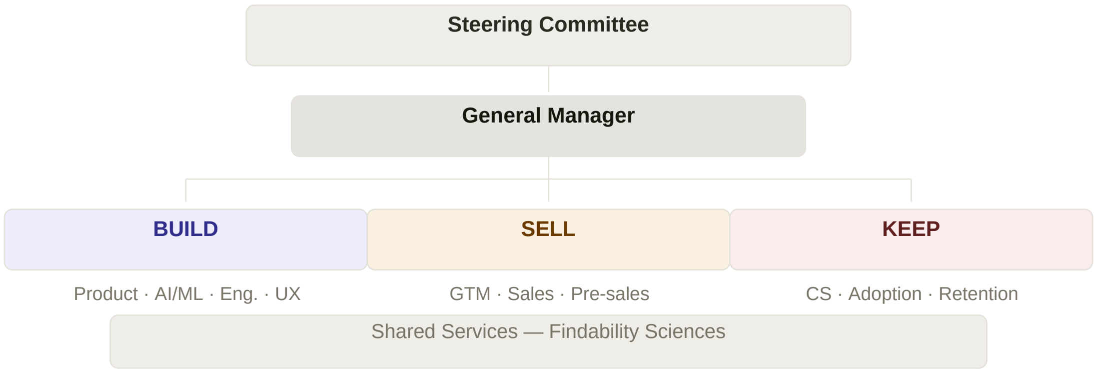
STOMA NEXUS

The prescriptive layer, answering: what needs to be done? Builds world models and digital twins of mill processes, applies optimisation solvers, and incorporates LLM-based copilots and agentic workflows to generate operating setpoint prescriptions. The most technically ambitious layer, owned jointly by AI/ML Engineer and GenAI/Agentic AI Engineer. Positioned as a co-development engagement with early accounts.

→ Commercial sequencing: land with Terra + Basic, then introduce Prime or Sense depending on client profile (industrial maintenance team = Prime first; agricultural focus = Sense first). Nexus follows when Prime adoption is solid. No layer should be sold before the previous one is genuinely adopted. CS & Adoption Lead holds veto authority.

4. Organisational structure

Stomata Labs is organised as a lean venture appropriate for early commercialisation. The three pillars sit at the same level under the General Manager, supported by Findability Sciences shared services.



Stomata Labs organisational structure.

4.1 Governance

The **Steering Committee** — composed of the CEOs of Pantaleon and Findability Sciences — holds strategic governance authority. It meets monthly and is responsible for budget approvals, partner commitment decisions, market expansion choices, and disputes that cannot be resolved operationally.

The **General Manager** is the single accountable executive. This role holds P&L; responsibility, delivery governance, resolves cross-pillar conflicts, manages commercial vendor agreements (including satellite data providers for Sense), and handles partner management. Without a clearly named GM, daily decisions escalate to a committee that meets monthly.

4.2 Build pillar

Build is responsible for everything the product is. The Head of Product & AI leads a nine-person team covering the full product and engineering capability required to deliver and maintain all five Stoma layers. The team is structured across two functional groups: **layer-facing technical roles** (AI/ML Engineer, GenAI/Agentic AI Engineer, Geospatial/RS Engineer, Domain Data Scientist, Data/Cloud Engineer) and **platform and delivery roles** (Full Stack Developer, MLOps/DevOps, UI/UX Designer, QA/Validation Engineer).

This structure reflects the reality of delivering a five-layer AI product in an industrial context. Each layer has distinct technical requirements that cannot be handled by a generalist pool.

4.3 Sell pillar

Sell owns the full commercial motion from pipeline generation through contract signature. The GTM Lead is supported by three Domain Solution Engineers (agriculture, industrial, logistics) who carry a dual accountability: 70% Sell activities, 30% product intelligence contribution. A formal layer-readiness framework governs which DSE can lead which type of conversation — including a new Sense readiness tier requiring geospatial and agronomy collaboration.

4.4 Keep pillar

Keep is responsible for everything post-signature. The CS & Adoption Lead carries a five-layer onboarding mandate: a distinct programme for each Stoma layer, including a Sense-specific programme covering satellite data onboarding, field calibration, vegetation and variability training, and insights review. At six active accounts, this role splits into Account Manager and Change Management & Training Specialist.

4.5 Shared services

Finance, legal, HR, DevOps/Platform Operations, QA, and security are provided by Findability Sciences. Service levels and cost allocation must be explicitly defined in the joint venture agreement. Client data isolation in the multi-client Terra data lake is a legal, ethical, and commercial requirement requiring Security and Legal sign-off before the first multi-client deployment.

5. Roles and responsibilities

Each role has a defined purpose and a set of owned outcomes.

General Manager
Governance

Single accountable executive. Owns P&L;, delivery governance, resolves cross-pillar conflicts, manages vendor agreements (incl. satellite providers for Sense), handles partner management. Prevents the venture from being governed by committee.

Head of Product & AI
Build

Defines what gets built across all five Stoma layers. Owns product roadmap, AI/ML methodology, platform architecture, data governance, and UX quality bar. The single person accountable for the product being technically sound and operationally credible.

AI / ML Engineer
Build

Designs and maintains predictive AI and anomaly detection models for Prime and Nexus. Covers signal processing, reconstruction error methods, and optimisation intelligence. A role, not a single headcount — can have multiple specialists under it as the venture scales.

GenAI / Agentic AI Engineer
Build

Integrates LLMs, copilots, and agentic workflow automation into Basic and Nexus. Owns the interface between generative AI and deterministic model outputs. Ensures GenAI features are production-safe before client deployment.

Geospatial / RS Engineer
Build

Owns Stoma Sense: satellite data integration, remote sensing model development, and field intelligence outputs. Works closely with Domain DS for agronomic validation. The specialist role that makes Sense possible — cannot be substituted by a generalist data engineer.

Domain Data Scientist
Build

The bridge between domain expertise and model development. Validates all model assumptions across all five layers against operational mill reality. Critical for KPI logic (Basic), anomaly plausibility (Prime), field index interpretation (Sense), and Nexus prescription safety.

Data / Cloud Engineer
Build

Owns Terra: data lake architecture, ingestion pipelines from all sources (IoT, satellite, climate APIs), and cloud platform management. Defines and enforces data quality SLAs. If Terra quality is insufficient for Prime or Sense, this must be surfaced before models are trained.

Full Stack Developer Build	Builds applications, APIs, dashboards, and workflow systems across all product layers. The engineering role that makes the product visible and usable — dashboards for Basic, workflow interfaces for Nexus, and field intelligence views for Sense.
MLOps / DevOps Build	Owns AI model deployment, monitoring, and lifecycle governance. Supports AI/ML and GenAI engineers in packaging models for production. Coordinates with Findability Sciences DevOps on underlying infrastructure. Ensures production model behaviour stays aligned with validation profiles.
UI / UX Designer Build	Designs operational workflows, visualisations, and interfaces across all product layers. For Nexus and Sense — high-stakes environments where display design directly affects operator decisions — works with Head of Product & AI in an active design direction mode rather than an execution mode.
QA / Validation Engineer Build	Owns test coverage, reliability standards, and production validation gates across all five layers. Designs domain-specific test cases with Domain DS. No release proceeds without QA sign-off. Coordinates with Findability Sciences QA on process alignment.
GTM Lead Sell	Owns the full commercial motion from pipeline to contract. Leads pricing strategy, the five-layer expansion commercial playbook, and coordinates Domain Solution Engineers. Single person accountable for new revenue.
Domain Solution Eng. ×3 Sell	Technical credibility engine of the sales cycle. Three engineers (agriculture, industrial, logistics) lead pre-sales at the level of mill engineers and process directors. Dual accountability: GTM Lead (commercial) and Head of Product & AI (product intelligence). Governed by 70/30 time split and a four-tier layer-readiness framework.
CS & Adoption Lead Keep	Owns everything post-signature: onboarding, change management, renewals, field feedback, and expansion coordination. Five-layer onboarding mandate including a dedicated Sense programme. Holds veto authority over expansion timing.

6. How the pillars work together

The three pillars are interdependent. Build produces what Sell promises and what Keep delivers. Sell generates commercial intelligence that shapes Build's roadmap. Keep provides field reality that validates and challenges both Build and Sell. The full protocol set — now covering all five product layers and all nine Build roles — is available in the interactive HTML guide.

6.1 Management interactions

Weekly: pipeline review with GTM Lead; priority alignment with Head of Product & AI (now covering all five layers). **Bi-weekly:** account health review with CS & Adoption Lead. **Monthly:** shared services SLA review (including satellite data provider SLAs for Sense); DSE time allocation review; venture health report to Steering Committee. **Quarterly:** roadmap strategic review by layer; pricing strategy review.

The GM's primary operational role is to resolve conflicts between pillars before they become blockers. The most common conflict at this stage is between Build capacity across five layers and commercial commitments across three pillars.

6.2 Build interactions

The Build team operates on a sprint cadence. The Head of Product & AI runs weekly backlog grooming across all nine roles, maintaining explicit capacity splits across all five layers. The most intensive internal collaboration is the weekly session between the Process Optimisation AI/ML Engineer and the Domain Data Scientist for Nexus model validation. The Geospatial/RS Engineer and Domain DS run a bi-weekly session for Sense output validation — an equally critical gate given the direct impact on agronomist decision-making.

Build receives external intelligence from two sources: Domain Solution Engineers (bi-weekly field-to-product debrief) and the CS & Adoption Lead (bi-weekly structured field feedback per layer). Both channels must be protected across all five layers.

6.3 GTM interactions

The GTM Lead runs a weekly pipeline review with the GM and bi-weekly alignment with the Head of Product & AI. A four-tier DSE layer-readiness framework governs opportunity assignment: Terra/Basic (all DSEs independently), Prime (technical preparation required), Nexus (HPA mandatory), and Sense (geospatial and agronomy collaboration with Geospatial/RS Engineer). On contract signature, a formal written handoff to CS & Adoption Lead is mandatory.

6.4 Customer Success interactions

The CS & Adoption Lead delivers structured field feedback to the Head of Product & AI bi-weekly, segmented across all five product layers. For Nexus accounts, CS operates a 90-day intensive adoption programme with the AI/ML Engineer and relevant DSE. For Sense accounts, CS operates a dedicated programme with the Geospatial/RS Engineer covering satellite data onboarding, field calibration, and vegetation intelligence training — a materially different journey from any other product layer.

CS holds veto authority over next-layer expansion. No expansion is proposed by GTM until CS confirms that the current layer is genuinely adopted.

7. Critical protocols

Nine protocols are designated as critical — interactions that must not be skipped, deprioritised, or treated as informal.

■ **Process model validation with Domain Data Scientist**

Scope: Build · Nexus

The only gate between a world model prescription and a mill operator acting on it. Must produce written validation outputs. If this becomes a casual check-in, an operationally unsafe setpoint can reach a client.

■ **Domain knowledge debrief — field to product**

Scope: Sell → Build · Cross

The mechanism by which product development stays connected to mill operational reality across all five layers. If cancelled when calendars fill, model assumptions drift and quality degrades invisibly.

■ **Terra data quality gate enforcement**

Scope: Build · Terra → Prime

Prime and Sense model performance depends on Terra data quality. This gate must produce a written quality report before any model is deployed. Skipping it under delivery pressure produces a degraded model.

■ **Mandatory Nexus scoping co-presence**

Scope: Sell · Nexus

No Nexus scoping conversation occurs without Head of Product & AI present. One incorrectly scoped engagement can result in a failed delivery in a market where reputation travels fast.

■ **CS layer adoption readiness confirmation**

Scope: Keep → Sell · Cross

CS holds veto authority over expansion timing. Expansion before adoption is the fastest path to churn.

■ **Commercial handoff on contract signature**

Scope: Sell → Keep · Cross

Everything promised during sales must transfer in writing to the person who will deliver it. One-page written handoff document is mandatory across all layers.

■ Nexus physical plausibility violation escalation

Scope: Build · Nexus

If Domain Data Scientist identifies a prescription that violates physical process constraints, all work on that branch stops immediately. No timeline or commercial pressure overrides this.

■ Client data isolation architecture sign-off

Scope: Build · Terra

Before any new client data enters the Terra data lake, Head of Product & AI must sign off on isolation architecture. Legal requirement — cannot be assumed.

■ Sense ground-truth calibration sign-off

Scope: Build + Keep · Sense

Before Sense outputs are shared with a client, Geospatial/RS Engineer and Domain DS must confirm that satellite-derived indices have been ground-truthed against actual field conditions for that geography. Unvalidated Sense outputs that contradict what agronomists observe in the field will destroy trust in the entire platform.

8. Setting up for OKRs

The organisational structure described in this document provides the natural framework for Stomata Labs' OKR process. OKRs should cascade across four levels: venture, pillar, role, and protocol.

OKR hierarchy

Venture level	Set by the GM with Steering Committee. Revenue targets, product layer delivery milestones across all five layers, retention and adoption rates. Reviewed quarterly.
Pillar level	One OKR set per pillar (Build, Sell, Keep). Each pillar objective directly supports a venture-level objective. Reviewed quarterly; key results tracked monthly.
Role level	Individual OKRs for each named role. Derived from pillar OKRs. Key results are measurable outputs of defined responsibilities. Reviewed quarterly; checked monthly.
Protocol level	The working protocols define operational excellence at the interaction level. Key results can reference protocol compliance — e.g. "DSE layer-readiness briefing delivered 100% of weeks" or "Sense calibration sign-off completed before every Sense client go-live".

Suggested OKR anchors by pillar

The following are indicative objective areas. Key results should be defined collaboratively with each pillar lead.

Pillar	Objective areas	Indicative key result types
Build	5-layer delivery cadence Model and Sense quality Platform reliability UX adoption	On-time delivery rate per layer Model accuracy + Sense calibration rate Terra/MLOps uptime SLA UI support tickets + feature adoption rate
Sell	Pipeline generation Deal conversion Layer expansion DSE intelligence quality	Qualified pipeline value Win rate by layer Layer expansion per account DSE debrief cadence and quality score
Keep	Account adoption (all 5 layers) Retention and renewal Field intelligence loop Sense and Nexus programmes	Active usage rate per layer Net Revenue Retention Field feedback sessions delivered Nexus + Sense adoption programme completion
GM / Venture	Revenue Product credibility Team stability Partner governance	ARR vs. plan Accounts at full-layer adoption Employee retention Shared services SLA compliance

→ OKRs should be set after this document has been reviewed by all team members. The most effective first OKR cycle is one where each person's key results are directly traceable to a working protocol. The Steering Committee should review and approve venture-level OKRs at the first quarterly review.

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